

Schneider EasyLogic PM2100 series

Modbus Register Map

PM2110 / PM2120 / PM2130 (METSEPM21xx) | User manual ref. NHA2779002-12

Protocol basics

Mode: Modbus RTU over RS-485. Default: address 1, 19200 baud, even parity. Up to 32 devices per bus.

Data types. FLOAT32 = IEEE-754 32-bit float across 2 registers (big-endian, high word first). INT64 = 4 registers. 4Q_FP_PF = Schneider four-quadrant float power-factor encoding (value -2 to +2, decode per the register-list References tab). DATETIME = 4-register Schneider date/time.

Addressing. Numbers below are 1-based register numbers (as published by Schneider). On the wire, protocol/PDU address = register number - 1 (e.g. register 3000 Current A is sent as 0x0BB7). Many masters expect the -1 offset; some add it automatically. Read with function 0x03.

Model note. PM2110 has pulse output only (no RS-485). PM2120/PM2130 have RS-485. Harmonics beyond THD and some advanced items are PM2130 only.

Real-time metering (1-second, FLOAT32)

Register	Regs	Data type	Unit	Description
3000	2	FLOAT32	A	Current A
3002	2	FLOAT32	A	Current B
3004	2	FLOAT32	A	Current C
3006	2	FLOAT32	A	Current N (neutral)
3010	2	FLOAT32	A	Current Average
3012	2	FLOAT32	%	Current Unbalance A
3018	2	FLOAT32	%	Current Unbalance Worst
3020	2	FLOAT32	V	Voltage A-B
3022	2	FLOAT32	V	Voltage B-C
3024	2	FLOAT32	V	Voltage C-A
3026	2	FLOAT32	V	Voltage L-L Average
3028	2	FLOAT32	V	Voltage A-N
3030	2	FLOAT32	V	Voltage B-N
3032	2	FLOAT32	V	Voltage C-N
3036	2	FLOAT32	V	Voltage L-N Average
3044	2	FLOAT32	%	Voltage Unbalance L-L Worst
3052	2	FLOAT32	%	Voltage Unbalance L-N Worst
3054	2	FLOAT32	kW	Active Power A
3056	2	FLOAT32	kW	Active Power B
3058	2	FLOAT32	kW	Active Power C
3060	2	FLOAT32	kW	Active Power Total
3062	2	FLOAT32	kVAR	Reactive Power A
3064	2	FLOAT32	kVAR	Reactive Power B
3066	2	FLOAT32	kVAR	Reactive Power C
3068	2	FLOAT32	kVAR	Reactive Power Total
3070	2	FLOAT32	kVA	Apparent Power A
3072	2	FLOAT32	kVA	Apparent Power B
3074	2	FLOAT32	kVA	Apparent Power C

Register	Regs	Data type	Unit	Description
3076	2	FLOAT32	kVA	Apparent Power Total
3078	2	4Q_FP_PF	---	Power Factor A
3080	2	4Q_FP_PF	---	Power Factor B
3082	2	4Q_FP_PF	---	Power Factor C
3084	2	4Q_FP_PF	---	Power Factor Total
3086	2	4Q_FP_PF	---	Displacement PF A
3092	2	4Q_FP_PF	---	Displacement PF Total
3110	2	FLOAT32	Hz	Frequency

Energy, floating point (FLOAT32, kWh / kVARh / kVAh)

Register	Regs	Data type	Unit	Description
2700	2	FLOAT32	kWh	Active Energy Delivered (into load)
2702	2	FLOAT32	kWh	Active Energy Received (out of load)
2704	2	FLOAT32	kWh	Active Energy Delivered + Received
2706	2	FLOAT32	kWh	Active Energy Delivered - Received
2708	2	FLOAT32	kVARh	Reactive Energy Delivered
2710	2	FLOAT32	kVARh	Reactive Energy Received
2712	2	FLOAT32	kVARh	Reactive Energy Delivered + Received
2714	2	FLOAT32	kVARh	Reactive Energy Delivered - Received
2716	2	FLOAT32	kVAh	Apparent Energy Delivered
2718	2	FLOAT32	kVAh	Apparent Energy Received
2720	2	FLOAT32	kVAh	Apparent Energy Delivered + Received
2722	2	FLOAT32	kVAh	Apparent Energy Delivered - Received
2732	2	FLOAT32	kVARh	Reactive Energy in Quadrant I
2734	2	FLOAT32	kVARh	Reactive Energy in Quadrant II
2738	2	FLOAT32	kVARh	Reactive Energy in Quadrant III
2740	2	FLOAT32	kVARh	Reactive Energy in Quadrant IV

Energy, integer accumulators (INT64, Wh / VARh / VAh)

Register	Regs	Data type	Unit	Description
3204	4	INT64	Wh	Active Energy Delivered (into load)
3208	4	INT64	Wh	Active Energy Received (out of load)
3212	4	INT64	Wh	Active Energy Delivered + Received
3216	4	INT64	Wh	Active Energy Delivered - Received
3220	4	INT64	VARh	Reactive Energy Delivered
3224	4	INT64	VARh	Reactive Energy Received
3236	4	INT64	VAh	Apparent Energy Delivered
3240	4	INT64	VAh	Apparent Energy Received
3200	4	DATETIME	---	Accumulated Energy Reset Date/Time

Demand (FLOAT32)

Register	Regs	Data type	Unit	Description
3764	2	FLOAT32	kW	Active Power: Last Demand
3766	2	FLOAT32	kW	Active Power: Present Demand

Register	Regs	Data type	Unit	Description
3768	2	FLOAT32	kW	Active Power: Predicted Demand
3770	2	FLOAT32	kW	Active Power: Peak Demand
3772	4	DATETIME	---	Active Power: Peak Demand Date/Time
3780	2	FLOAT32	kVAR	Reactive Power: Last Demand
3786	2	FLOAT32	kVAR	Reactive Power: Peak Demand
3796	2	FLOAT32	kVA	Apparent Power: Last Demand
3802	2	FLOAT32	kVA	Apparent Power: Peak Demand
3876	2	FLOAT32	A	Current: Last Demand (avg)
3882	2	FLOAT32	A	Current: Peak Demand (avg)

Total harmonic distortion (FLOAT32, %)

Register	Regs	Data type	Unit	Description
21300	2	FLOAT32	%	THD Current A
21302	2	FLOAT32	%	THD Current B
21304	2	FLOAT32	%	THD Current C
21322	2	FLOAT32	%	THD Voltage A-B
21324	2	FLOAT32	%	THD Voltage B-C
21326	2	FLOAT32	%	THD Voltage C-A
21330	2	FLOAT32	%	THD Voltage A-N
21332	2	FLOAT32	%	THD Voltage B-N
21334	2	FLOAT32	%	THD Voltage C-N

Individual harmonic magnitudes (H1–H31, PM2130) begin at register 21712 for Voltage A-B, then 22100 for Voltage B-C, and continue in per-input blocks; each harmonic is FLOAT32 (2 registers) spaced 6 registers apart. See the full register list for the complete harmonic block.

Key setup registers (write via command interface)

Register	Regs	Data type	Unit	Description
2014	1	INT16U	---	Number of Phases
2016	1	INT16U	---	Power System Configuration
2017	1	INT16U	Hz	Nominal Frequency
2018	2	FLOAT32	V	Nominal Voltage
2020	2	FLOAT32	A	Nominal Current
2026	2	FLOAT32	V	VT Primary
2028	1	INT16U	V	VT Secondary
2030	1	INT16U	A	CT Primary
2031	1	INT16U	A	CT Secondary
2036	1	INT16U	---	VT Connection Type

Important: the PM2100 is NOT a simple write-anywhere map. CT/VT/system changes go through the Protected Command Interface (registers 5000–5127). Acquire the command semaphore by reading register 5001, then write the Basic Meter Setup command (2000) with the full parameter block plus the semaphore, and confirm via the Command Result registers 5125–5127 (result in 5126 must be 0). Do not write setup registers directly.

Source: Schneider Electric “EasyLogic PM2100 / PM2200 / EM6400NG Register List” (Public EM6400_PM2xxx PMC Register List). This map is shared across those families. The user manual NHA2779002-12 itself contains no register table; it directs you to this register list. Download the current version from se.com (search “PM2100 PM2200 Register List”) and verify against your firmware before commissioning.